REACTIONS AND EQUATIONS

The Law of Conservation of Mass states that

Why is this important to remember when we are reacting two substances together?

Balancing Equations

Because of the Law of Conservation of Mass, all of the atoms that appear on the reactant side of the equation must also appear on the product side.

Let us see what balancing involves.

Consider the following equation:

$NaOH + Pb(NO_3)_2 \rightarrow Pb(OH)_2 + NaNO_3$

In this example, the numbers on each side do not match.

Let's try using a T diagram

	Reactants	Products
Na	1	1
OH	1	2
Pb	1	1
NO_3	2	1

You will notice that we didn't split the OH or the NO_3 into separate atoms. This is because the OH stays as OH on both sides and NO_3 stays as NO_3 on both sides. It makes it simpler for us to balance to leave them as they are.

Balancing involves inserting numbers in front of formulae so that the atoms can be equal on both side.

In this example, if we look at the T diagram, you will see that to start with, the sodium is equal on both sides so we will leave this one for now.

Next you will see that the OH is different. We will have to make them the same. So if we put a 2 in front of the NaOH, we can change the number of OHs on the reactant side.

2NaOH + $Pb(NO_3)_2$ → $Pb(OH)_2$ + $NaNO_3$

This changes our T diagram

	Reactants	Products
Na	~~1~~ 2	1
OH	~~1~~ 2	2
Pb	1	1
NO_3	2	1

You will notice that the sodium has also changed. This is because we have 2 NaOH so now we have 2 Na and 2 OH.

If we keep working down the diagram, the next one that is out of balance is the NO_3. Let's put a 2 in front of the $NaNO_3$ to balance this one.

2NaOH + $Pb(NO_3)_2$ → $Pb(OH)_2$ + 2$NaNO_3$

	Reactants	Products
Na	~~1~~ 2	~~1~~ 2
OH	~~1~~ 2	2
Pb	1	1
NO_3	2	~~1~~ 2

If we look closely at our T diagram, you will see that the numbers in the reactants column is the same as the products column. This means that our equation is now balanced.

Try this one as a class

$NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$

REMEMBER
There must be the same number of atoms on each of the equation.
You cannot change a formula to help you balance.
You can only put numbers IN FRONT of a substance to change the number of atoms.

Try to balance these equations.

1) $Mg + HCl \rightarrow MgCl_2 + H_2$
2) $Mg + O_2 \rightarrow MgO$
3) $SO_2 + O_2 \rightarrow SO_3$
4) $KClO_3 \rightarrow KCl + O_2$

All we need now is plenty of practice.

a) $NO + O_2 \rightarrow NO_2$

b) $Fe + HCl \rightarrow FeCl_2 + H_2$

c) $Al + O_2 \rightarrow Al_2O_3$

d) $FeS + HCl \rightarrow FeCl_2 + H_2S$

e) $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$

f) $Mg + CO_2 \rightarrow MgO + C$

g) $Ca(OH)_2 + CH_3COOH \rightarrow Ca(CH_3COO)_2 + H_2O$

h) $KOH + H_2SO_4 \rightarrow K_2SO_4 + H_2O$

i) $Pb(NO_3)_2 + ZnCl_2 \rightarrow Zn(NO_3)_2 + PbCl_2$

j) $AlCl_3 + NaOH \rightarrow Al(OH)_3 + NaCl$

k) $Ca + H_2O \rightarrow Ca(OH)_2 + H_2$

l) $Ca(NO_3)_2 + (NH_4)_2CO_3 \rightarrow CaCO_3 + NH_4NO_3$

m) $K_3PO_4 + Fe(NO_3)_2 \rightarrow Fe_3(PO_4)_2 + KNO_3$

Here are some more challenging examples.

1) $C_{12}H_{22}O_{11} + O_2 \rightarrow CO_2 + H_2O$

2) $Mg(HCO_3)_2 \rightarrow MgO + CO_2 + H_2O$

3) $MgCO_3 + H_3PO_4 \rightarrow Mg_3(PO_4)_2 + H_2O + CO_2$

4) $S + Fe_2O_3 \rightarrow Fe + SO_2$

5) $Cu + HNO_3 \rightarrow Cu(NO_3)_2 + NO_2 + H_2O$

6) $MnO_2 + HCl \rightarrow MnCl_2 + H_2O + Cl_2$

7) $FeS + O_2 \rightarrow Fe_2O_3 + SO_2$

8) $CH_3OH + O_2 \rightarrow H_2O + CO_2$

9) $NH_3 + O_2 \rightarrow N_2 + H_2O$

10) $NO_2 + H_2O \rightarrow HNO_3 + NO$

11) $KMnO_4 + HCl \rightarrow MnCl_2 + KCl + H_2O + Cl_2$

Activity: What do students find the hardest thing to believe about chemistry.

Balance these equations by writing in all the coefficients, including ‘1’. For each equation, find its sequence of coefficients in the key list and write the corresponding letter next to the number of the equation. Read down the letters to find the answer.

$Ba(NO_3)_2 + K_2SO_4 \rightarrow KNO_3 + BaSO_4$

$CH_4 + O_2 \rightarrow H_2O + CO_2$

$CuO + HNO_3 \rightarrow Cu(NO_3)_2 + H_2O$

$Fe + Al_2O_3 \rightarrow Fe_2O_3 + Al$

$HCl + CaO \rightarrow CaCl_2 + H_2O$

$Al_2O_3 + C \rightarrow CO + Al$

$FeCl_3 + AgNO_3 \rightarrow Fe(NO_3)_3 + AgCl$

$NaOH + H_3PO_4 \rightarrow Na_3PO_4 + H_2O$

$Fe_2O_3 + C \rightarrow Fe + CO$

$Pb(NO_3)_2 + Na_2SO_4 \rightarrow NaNO_3 + PbSO_4$

$NaOH + HCl \rightarrow NaCl + H_2O$

$Na + H_2O \rightarrow H_2 + NaOH$

$H_2SO_4 + Cu(OH)_2 \rightarrow CuSO_4 + H_2O$

$BaO + HNO_3 \rightarrow Ba(NO_3)_2 + H_2O$

$Pb(NO_3)_2 + KCl \rightarrow PbCl_2 + KNO_3$

$Al(OH)_3 + HNO_3 \rightarrow H_2O + Al(NO_3)_3$

$H_2O + K \rightarrow H_2 + KOH$

KEY LIST:

A:	1,	1,	1,	1	F:	1,	2,	1,	2	U:	1,	3,	3,	1	W:	2,	1,	2,	1
B:	1,	1,	1,	2	H:	1,	2,	2,	1	S:	1,	3,	3,	2	N:	2,	2,	1,	2
C:	1,	1,	2,	1	T:	1,	3,	1,	3	I:	2,	1,	1,	1	G:	3,	1,	1,	2
E:	1,	2,	1,	1	Y:	1,	3,	2,	3	M:	2,	1,	1,	2	R:	3,	1,	1,	3